THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A homogeneous mixture of two or more substances is called a: 1

(a) solution **(b)** suspension **(c)** compound **(d)** element

2. Which of these solutions is a solid in a solid? 1

(a) soda water **(b)** air **(c)** brass **(d)** vinegar

3. The mass percentage of a solution is: 1

(a) mass of solute x mass of solvent
(b) (mass of solvent / mass of solute) x 100
(c) (mass of solute / mass of solution) x 100
(d) (mass of solution / mass of solute) x 100

4. In a solution of sugar in water, sugar is the: 1

(a) precipitate **(b)** solution **(c)** solute **(d)** solvent

5. In air, the solvent is: 1

(a) nitrogen **(b)** argon **(c)** carbon dioxide **(d)** oxygen

6. 20 g of salt is dissolved in 80 g of water. The mass percentage of salt is: 1

(a) 20 per cent **(b)** 25 per cent **(c)** 80 per cent **(d)** 10 per cent

7. In a solution of sugar in water, water is the: 1

(a) alloy **(b)** suspension **(c)** solvent **(d)** solute

8. A mixture in which particles settle down on standing is a: 1

(a) suspension **(b)** colloid **(c)** solution **(d)** compound

9. 10 g of sugar is dissolved in 90 g of water. The mass of solution is: 1

(a) 80 g **(b)** 100 g **(c)** 90 g **(d)** 10 g

10. Water is called the universal solvent because it: 1

(a) dissolves a large number of substances **(b)** boils at 100 degrees Celsius
(c) dissolves all substances **(d)** is colourless

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define solute, solvent and solution. 2

12. State two differences between a solution and a suspension. 2

13. Find the mass percentage of a solution with 5 g salt in 45 g water. 2

14. What is meant by the solubility of a substance? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show that the solubility of a solid increases with temperature. 3

16. How much solute is needed to prepare 200 g of a 10 per cent salt solution? 3

17. State three uses of solutions in daily life. 3

OR

Explain how you prepare a saturated solution of common salt. 3

18. Explain why a cold drink in a warm room goes flat sooner. 3

OR

Explain why oil does not dissolve in water but dissolves in kerosene. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the characteristics of solutions, suspensions and colloids in a table of five points with examples. 5

OR

Describe the types of solutions on the basis of the physical states of solute and solvent with an example each. 5

20. Explain how temperature affects the solubility of solids and gases with two examples each. 5

OR

Explain the Tyndall effect and use it to distinguish a solution from a colloid. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND